

The following specification describe the HL7 order import via the ORM message.

I. Transaction Structure

Name	Description	Comments
MSH	Message Header	This segment should always be present in the ORM message
PID	Patient Identification	There should be a PID segment to give the patient demographic information
GT1	Guarantor Info.	
IN1	Insurance Info	Up to 2 IN1 records for primary and secondary insurance may be sent
DG1	Diagnosis Info	Multiple DG1 may be sent
RXO	Patient Medications	Patient's Medications are necessary for toxicology orders
ORC	Common order info	Used to send ordering provider
OBR	Order Info	
OBX	Ask Order Entry Questions	

II. Message Delimiters

Character Function	Character
Segment Terminator	<CR> LF
Field Separator	(pipe)
Component Separator	^ (caret)
Sub-Component Separator	& (ampersand)
Repetition Separator	~ (tilde)
Escape Character	\ (back slash)

III Conventions

Character Function	Character
Required field this should always be sent	R
Conditional field	C
Optional field this info will be processed if available	O
Not Supported	N

MSH-- MESSAGE HEADER SEGMENT

The Message Header (MSH) segment defines the intent, source, destination, and some specifics of the syntax of a message.

Field Name	Seg. Name	Comment	Len	R/O
Segment Type ID	MSH.00	MSH	3	R
Encoding Characters	MSH.01	^~\&	4	R
Sending Application	MSH.02	Sending Application ()	30	R
Sending Facility	MSH.03	Sending Facility (Account#)	30	R
Receiving Application	MSH.04	COMTRON	30	O
Receiving Facility	MSH.05	Receiving Facility (Lab Name)	30	O
Message Date & Time	MSH.06	Format: YYYYMMDDHHMMSS	12	R
Security	MSH.07	Not Supported		N
Message Type	MSH.08	ORM^O01		R
Message Control ID	MSH.09	A number that uniquely identifies the message		R
Processing ID	MSH.10	P-Production T-Test	1	R
Version ID	MSH.11	2.3	8	R

SAMPLE OUTPUT

MSH|^~\&|ABC|4009|COMTRON|ACUPATH|200804081640||ORM^O01|0000005723|P|2.3||||

PID PATIENT IDENTIFICATION

The Patient Identifier (PID) segment is used by all applications as the primary means of communicating patient identification information. This segment contains permanent patient identifying and demographic information that, for the most part, is not likely to change frequently.

Field Name	Seg. Name	Comment	Len	R/O
Segment Type ID	PID.00	PID	3	R
Sequence Number	PID.01	Serial Number starting from 1	4	R
Patient ID (Internal ID)	PID.02	Patient ID (Client's Patient Chart ID)	20	R
Patient ID (External ID)	PID.03	External ID (NOT USED)		N
Alternate Patient ID	PID.04	External MRN#		N
Patient Name	PID.05	Last^First^MI	30	R
Mother's Maiden Name	PID.06	Not Supported		
Date of Birth	PID.07	YYYYMMDD	28	R
Sex	PID.08	F Female M Male U Unknown	1	O
Patient Alias	PID.09	Not Supported		N
Race	PID.10	Not Supported		O
Patient Address	PID.11	Addr1^Addr2^City^State^Zip	106	O
County Code	PID.12	Not Supported		N
Phone # - Home	PID.13	NNNNNNNNNN	10	O
Phone # - Business	PID.14	Not Supported		N
Primary Language	PID.15	Not Supported		N
Marital Status	PID.16	Not Supported		N
Religion	PID.17	Not Supported		N
Patient Account Number	PID.18	Not Supported		N
SSN# - Patient	PID.19	NNNNNNNNNN	9	O

SAMPLE OUTPUT

PID|2|123456789||SMITH^JOHN^|19560114|M||20 MAIN STREET^BLACKWOOD^NJ^08012|

Field Mnemonic	Use/Value Field	Required Fields	Max Size	Delimiter
GT1-0	Segment Type ID • Always “GT1”	R	3	()
GT1-1	Transmission Sequence Number • First is "1", second is "2", etc.	R	4	()
GT1-2	Unused	N		()
GT1-3	Guarantor Name			
	• Responsible Party Last Name	R	20	(^)
	• Responsible Party First Name	R	20	(^)
	• Responsible Party Middle Initial	R	1	(^)
	Unused	N		(^)
	Unused	N		(^)
GT1-4	Unused	N		()
	Guarantor Address			
	• Address	R	35	(^)
	Unused	N		(^)
	• City	R	25	(^)
GT1-5	• State or Province	R	2	(^)
	• Zip or Postal Code	R	10	()
GT1-6	Guarantor Home Phone Number (use Guarantor or Patient) Format is “999-999-9999”	O	12	()
GT1-7	Unused	N		()
GT1-8	Unused	N		()
GT1-9	Unused	N		()
GT1-10	Unused	N		()
GT1-11	Guarantor Relationship to Patient • Valid values: “1” = Self “2” = Spouse “3” = Son “4” = Daughter “5” = Other	R	1	()
GT1-12	Guarantor ID Number (SSN) • Format is 999-99-9999	O	11	(CR)

Note:

The GT1 segment is required for Patient Billed demographic data sets. Although optional for other billing types, it is recommended that a GT1 data segment be provided with all demographic data sets, even if the patient is the guarantor.

GT1-13 – GT1-20 are unused. Placeholder delimiters are not required.

Field Mnemonic	Use/Value Field	Required Fields	Max Size	Delimiter
IN1-0	Segment Type ID • Always “IN1”	R	3	()
IN1-1	Transmission Sequence Number First is "1", second is "2", etc. <u>Insurance information should be sent in order of Coordination of Benefits Rank.</u>	R	4	()
IN1-2	Not Used	N	60	()
IN1-3	Insurance Company ID • Lab's Insurance Company Identification	R	59	()
IN1-4	Insurance Company Name	R	130	()
IN1-5	Insurance Company Address • Conditionally required for Carriers that are considered "Multi-Plan" Payors.			
	• Insurance Address 1	C	30	(^)
	• Insurance Address 2	C	30	(^)
	• City	C	16	(^)
	• State or Province	C	2	(^)
	• Zip or Postal Code	C	10	(^)
	Unused	N		()
IN1-6	Unused	N		()
IN1-7	Insurance Company Phone Number Format is “999-999-9999”	O	40	()
IN1-8	Group Number of Insured Patient • Conditionally required based upon individual payor contract requirements	C	19	()
IN1-9	Group Name	O	30	()
IN1-10	Unused	N		()
IN1-11	Unused	N		()
IN1-12	Unused	N		()
IN1-13	Unused	N		()
IN1-14	Unused	N		()
IN1-15	Unused	N	1	()
IN1-16	Name of Insured	R		
	• Last Name	R	20	(^)
	• First Name	R	20	(^)
	• Middle Initial	R	1	(^)
	Unused	N		(^)
	Unused	N		(^)
	Unused	N		()
IN1-17	Insured's Relationship to Patient • Valid HL7 values: “1” = Self “2” = Spouse “3” = Son “4” = Daughter “5” = Other	R	2	()
IN1-18	Unused	N		()
IN1-19	Insured's Address			
	• Address 1	R	35	(^)
	Unused	N		(^)
	• City	R	25	(^)
	• State or Province	R	2	(^)
	• Zip or Postal Code	R	10	(^)
	Unused	N		()

Field Mnemonic	Use/Value Field	Required Fields	Max Size	Delimiter
IN1-20	Assignment of Benefits • This field indicates whether the insured has assigned insurance benefits to the health care provider. • Either Y or N	O	2	()
IN1-21	Coordination of Benefits • The field indicates whether this insurance provides coverage in conjunction with or independent of other coverage the patient may have. • Y- Medigap • N- Not Medigap	O	2	()
IN1-22	Primary Payer "Y" = This insurance is the primary payer. "N" = This insurance is the secondary payer. "U" = Payer priority is unknown.	O	2	()
IN1-23	Unused	N		()
IN1-24	Unused	N		()
IN1-25	Unused	N		()
IN1-26	Unused	N		()
IN1-27	Unused	N		()
IN1-28	Unused	N		()
IN1-29	Unused	N		()
IN1-30	Unused	N		()
IN1-31	Type of Agreement • Conditionally required based upon the Billing contract with the carrier. • Workman's Compensation Flag (Y/N)	C	2	()
IN1-32	Unused	N		()
IN1-33	Unused	N		()
IN1-34	Unused	N		()
IN1-35	Unused	N		()
IN1-36	Policy Number (Insurance Number)	R	30	(CR)

Note:

IN1-37 – IN1-49 are unused. Placeholder delimiters are not required.

ORC COMMON ORDER SEGMENT

The Common Order (ORC) segment is used to send the requesting provider.

Field Name	Seg. Name	Comment	Len	R/O
Segment Type ID	ORC.00	ORC	3	R
Order Control	ORC.01	Not Supported	4	N
Place Order Number	ORC.02	Customer Specific order number	10	R
Filler Order Number	ORC.03	Lab Accession# (not used)		N
Placer Group Number	ORC.04	Not Supported		N
Order Status	ORC.05	Not Supported		N
Response Flag	ORC.06	Not Supported		N
Quantity/Timing	ORC.07	Not Supported		N
Parent	ORC.08	Not Supported		N
Date/Time of Transaction	ORC.09	YYYYMMDD	26	R
Entered By	ORC.10	Not Supported		N
Verified By	ORC.11	Not Supported		N
Ordering Provider	ORC.12	Doctor ID^LastName^FirstName^Num^IDType IDType: U= UPIN N=NPI	80	R

SAMPLE OUTPUT

ORC|RE||5276628 |||||200804081127||1770631822^ROBERTS^BARBARA^ 0^N

RXO Medication

The RXO segment holds information about medications that the patient is taking and it is optional. the specific results of the patient's test and it is required.

Field	Name	R	Notes
RXO 1.1	Identifier	O	Preferred identifiers are RXNorm
RXO 1.2	Medication Name	R	

SAMPLE OUTPUT

RXO|Plavix^Clopidogrel|

RXO| 352884^Percocet|

OBR OBSERVATION REQUEST SEGMENT

At least one OBR segment should be present for each Order Code associated with any PID segment. This segment is mandatory in ORM messages.

Field Name	Seg.Name	Field Comment	Len	R/O
Segment Type ID	OBR.00	OBR	3	R
Sequence No	OBR.01	Serial Number Starting from 1	4	R
Placer Order Number	OBR.02	Customer Specific order#	10	R
Filler Order Number	OBR.03	Lab Accession# (not used)		R
Observation Battery Identifier	OBR.04	Comtron's Test Code^Test Name	200	R
Priority	OBR.05	Not Supported		N
Requested Date/Time	OBR.06	Not Supported		N
Observation/Specimen Collection Date/Time #	OBR.07	Collection Date/Time #	26	R
Observation End Date/Time #	OBR.08	Not Supported		N
Collection Volume	OBR.09	Volume		O
Collector Identifier	OBR.10	Not Supported		N
Specimen Action Code	OBR.11	Not Supported		N
Danger Code	OBR.12	Not Supported		N
Relevant Clinical Information	OBR.13	Not Supported		N
Order Date/Time	OBR.14	Order Date/Time		R
Specimen Source	OBR.15	Not Supported		N
Ordering Provider	OBR.16	Doctor ID^Last Name^First Name^Num^^^^IDType	80	R
Ordering Provider's Phone Number	OBR.17	Not Supported		N
Alternate Specimen ID	OBR.18	Not Supported	40	N
Lab Info	OBR.19	Not Supported	40	N
Date/Time Observations Reported	OBR.22	Not Supported		
Result Status	OBR.25	Not Supported	1	O
Priority	OBR.27	^^^^^Priority R- Regular S- Stat	1	O
Link to parent order	OBR.29	Customers order number	80	O

SAMPLE OUTPUT

OBR|1|0905120003||0080379^VITAMIN D-25
HYDROXY^L||200905121000||^ML||N||1346280336^GUDZ^ALEXANDER^^^^N|(973)762-
6033|000959A||||||^R

OBX OBSERVATION/RESULT SEGMENT

This segment is used to send Ask at order entry question and answers.

It is an optional segment

Field Name	Seq. Nqme	Comment	Len	R/O
Segment Type ID	OBX.00	OBX	3	R
Sequence Number	OBX.01	Serial Number starting from 1	4	R
Type Value	OBX.02	ST- String Data	2	R
Observation Identifier	OBX.03	Question Code^Question^	60	R
Observation Sub-ID	OBX.04	Not Supported		N
Observation Value	OBX.05	Answer	28	R
Units	OBX.06	Not Supported	15	N
Reference Ranges	OBX.07	Not Supported	20	N
Abnormal Flags	OBX.08	Not Supported	2	N
Probability	OBX.09	Not Supported		N
Nature of Abnormal tests	OBX.10	Not Supported		N
Observation Result Status	OBX.11	Not Supported	1	O
Date Last Observation Normal Values	OBX.12	Not Supported		N
User Defined Access Checks	OBX.13	Not Supported		N
Date/Time of the Observation	OBX.14	Not Supported	26	N
Producer's ID	OBX.15	Not Supported	60	O
Producer's Address	OBX.16	Not Supported		O

Sample Output

OBX|1|ST|Q204^LMP^L||01/01/2000|

NTE NOTES AND COMMENTS

The Notes and Comments (NTE) segment contains notes and comments for the lab results and it is an optional segment.

Field Name	Seg. Name	Comment	Len	R/O
Segment Type ID	NTE.00	NTE	3	R
Sequence Number	NTE.01	Serial Number starting from 1	4	R
Source of Comment	NTE.02	Source of Comment	8	O
Comment	NTE.03	Comment	255	R

SAMPLE OUTPUT

NTE|1|L|NON-FASTING

SAMPLE SEGMENT:

MSH|^~^&|ABC CLINIC|13392|COMTRON||200805121446||ORM^O01|5|P|2.3|5|||
PID|1|123456|||PATIENT^TEST^||20000101|M||123 MAIN STREET, ^123 MAIN STREET,
^BROOKLYN^NY^11230||(516)555-1212||||13392^^^C|||||||
IN1|1||MED|EMPIRE MEDICARE SERVICES|P.O. BOX
69201^^HARRISBURG^PA^17106|||||||PI|PATIENT^TEST^|1|||||||N||||141249924A|||||||
IN1|2|HORIZON|HORIPOBOX000004|HORIZON NJ DIRECT|PO BOX 820^^NEWARK^NJ^07101-0820||21-
93500|||||PI|PATIENT^TEST^|1|||||||N||||NJX3HZN86482110|||||||
GT1|1||PATIENT^TEST^||34 SPRINGFIELD AVE^^HASBROUCK HGTS^NJ^07604||||1|||||
DG1|1|I9|427.31|Atrial Fibrillation|20061117000000|||||||
RXO|Plavix^Clopidogrel|
RXO|352884^Percocet|
ORC|NW|0905120003|||||200905121440||1346280336^GUDZ^ALEXANDER^^^^N|||||||
OBR|1|0905120003||0080379^VITAMIN D-25
HYDROXY^L||200905121000||^ML||N||||1346280336^GUDZ^ALEXANDER^^^^N|(973)762-
6033|000959A||||||^R
ORC|NW|0905120003|||||200905121440||1346280336^GUDZ^ALEXANDER^^^^N|||||||
OBR|2|0905120003||0080385^VITAMIN_D,1,25_DIHYDROXY^L||200905121000||^ML||N||||1346280336^GUDZ^ALE
XANDER^^^^N|(973)762-6033|000959A |||||^^R